

Locally Convex Topological Vector Spaces

Definitions and Examples

Definition: A topological vector space (TVS) is a vector space X with a topology $\mathcal{T}$ such that, with respect to this topology, the maps

- (i) $X \times X \rightarrow X, (x, y) \mapsto x + y$
- (ii) $\mathbb{F} \times X \rightarrow X, (\alpha, x) \mapsto \alpha x$

are continuous with respect to $\mathcal{T}$.

Example: If $\mathcal{P}$ is a family of seminorms on X , and $\mathcal{T}$ is the topology on X with subbasic sets $U(x_0, p, \varepsilon) = \{x \in X : p(x - x_0) < \varepsilon\}$, then $(X, \mathcal{T})$ is a topological vector space.

Definition: A locally convex topological vector space (LCS) is a TVS equipped with a family of seminorms $\mathcal{P}$ that satisfy

$$\bigcap_{p \in \mathcal{P}} \{x : p(x) = 0\} = \{0\}.$$

That is, the family $\mathcal{P}$ separates the points of X .

Proposition: Let X be a TVS, and let p be a seminorm on X . The following are equivalent:

- (a) p is continuous;
- (b) $\{x \in X : p(x) < 1\}$ is open;
- (c) $0 \in \{x \in X : p(x) < 1\}^\circ$;
- (d) $0 \in \{x \in X : p(x) \leq 1\}^\circ$;
- (e) p is continuous at 0;
- (f) there is a continuous seminorm q on X such that $p \leq q$.

Proof. We see immediately that (a) implies (b) since the set is equal to the inverse image of the open sub-set $[0, 1) \subseteq [0, \infty)$. Similarly, if $p^{-1}([0, 1))$ is open, then it is equal to its own interior, and since $0 \in p^{-1}([0, 1))$, it follows that there is a subbasic open neighborhood of 0 contained in $p^{-1}([0, 1))$, giving (c). Similarly, we obtain (d) by the fact that $0 \in p^{-1}([0, 1))^\circ \subseteq p^{-1}([0, 1])^\circ$.

Next, we see that (d) implies that for every $\varepsilon > 0$, $0 \in p^{-1}([0, \varepsilon])^\circ$, so if $(x_i)_i$ is a net in X converging to 0 and $\varepsilon > 0$, there is some i_0 such that for all $i \geq i_0$, $x_i \in \{x : p(x) \leq \varepsilon\}$. That is, $p(x_i) \leq \varepsilon$ for all $i \geq i_0$, whence p is continuous at 0. This gives (e).

If p is continuous at 0, then if $x_i \rightarrow x$ is a convergent net in X , the reverse triangle inequality gives

$$|p(x) - p(x_i)| \leq p(x_i - x),$$

whence $p(x_i) \rightarrow p(x)$, so p is continuous. This gives that (e) implies (a).

Taking $q = p$ yields (a) implies (f). We will show that (f) implies (e). If $x_i \rightarrow 0$ in X , then $q(x_i) \rightarrow 0$, but since $0 \leq p(x_i) \leq q(x_i)$, it follows that $p(x_i) \rightarrow 0$. $\square$

Example:

- (a) Let X be a completely regular topological space, and let $C(X)$ be the set of all continuous functions from X to $\mathbb{F}$. If K is compact in X , we may define a seminorm $p_K(f) = \sup_{x \in K} |f(x)|$. Then,

$$\{p_K : K \subseteq X \text{ compact}\}$$

forms a separating family of seminorms for $C(X)$.

- (b) Let G be an open subset of $\mathbb{C}$. If $H(G)$ is the set of holomorphic functions on G , then the same semi-

norms $\{p_K : K \subseteq G \text{ compact}\}$ forms a locally compact space whose topology is uniform convergence on compact subsets.

- (c) Let X be a normed vector space. For each $\varphi \in X^*$, let $p_\varphi(x) = |\varphi(x)|$. Then, $\mathcal{P} = \{p_\varphi : \varphi \in X^*\}$ forms a separating family of seminorms for X . The topology these seminorms induce is called the *weak topology* on X . It is often denoted $\sigma(X, X^*)$.
- (d) If X is a normed space, then for each $x \in X$, we may define a seminorm $p_x : X^* \rightarrow [0, \infty)$ by taking $p_x(\varphi) = |\varphi(x)|$. This defines a separating family of seminorms $\mathcal{P} = \{p_x : x \in X\}$, and the topology on X^* induced by these seminorms is called the *weak**, or w^* , topology. It is often denoted $\sigma(X^*, X)$.

Next, we focus on an examination of the “convexity” part of the notion of locally convex topological vector spaces.

Definition: A subset $A \subseteq X$ is said to be convex if for all $v, w \in A$, the segment

$$[v, w] := \{tv + (1-t)w : 0 \leq t \leq 1\}$$

is contained in A .

Note that the intersection of any family of convex subsets is convex.

Definition: If $A \subseteq X$ is any subset, then the *convex hull* of A , denoted $\text{conv}(A)$, is the intersection of all the convex subsets that contain A . If X is a TVS, the *closed convex hull* of A , denoted $\overline{\text{conv}}(A)$, is the intersection of all the closed convex sets that contain A .

Given any linear map $T : X \rightarrow Y$ of real vector spaces, and $C \subseteq X$ a convex subset of Y , then $T^{-1}(C)$ is convex in X . If X^* is the dual of any complex vector space, then since X is also a real vector space, it then follows that the sets

$$\begin{aligned} A_0 &= \{x : |f(x)| \leq 1\} \\ A_1 &= \{x : \text{Re}(f(x)) \leq 1\} \\ A_3 &= \{x : \text{Re}(f(x)) > 1\} \end{aligned}$$

are all convex.

Proposition: Let X be a TVS, and let A be a convex subset of X . Then,

- (a) $\overline{A}$ is convex;
- (b) if $a \in A^\circ$ and $b \in \overline{A}$, then

$$[a, b] := \{tb + (1-t)a : 0 \leq t < 1\}$$

is contained in A° .

Proof. Let $a \in A$, $b \in \overline{A}$, and let $0 \leq t \leq 1$. Let $(x_i)_i$ be a net in A such that $x_i \rightarrow b$. Then, $tx_i + (1-t)a \rightarrow tb + (1-t)a$. In particular, this means that if $b \in \overline{A}$ and $a \in A$, then $[a, b] \subseteq \overline{A}$. So, given any $a, b \in \overline{A}$, we have that the net $(x_i)_i \rightarrow b$ is such that $[x_i, a] \subseteq \overline{A}$, so it follows that $[b, a] \subseteq \overline{A}$, whence $\overline{A}$ is convex.

For (b), fix $0 < t < 1$, and set $c = tb + (1-t)a$ for some $a \in A^\circ$ and $b \in \overline{A}$. Since $a \in A^\circ$, there is an open neighborhood V of 0 such that $a + V \subseteq A$. Therefore, for any $d \in A$, we have

$$\begin{aligned} A &\supseteq td + (1-t)(a + V) \\ &= t(d - b) + tb + (1-t)(a + V) \\ &= (t(d - b) + (1-t)V) + c. \end{aligned}$$

We hope to find some $d \in A$ such that $0 \in t(d - b) + (1-t)V$, which we call U . The preceding inclusion then shows that $c \in A^\circ$, since U is an open neighborhood of 0 with $c + U \subseteq A$.

Finding such a d is equivalent to finding d such that $0 \in \frac{1}{t}(1-t)V + (d - b)$, or $d \in b - \frac{1}{t}(1-t)V$. Yet, we know that $0 \in -\frac{1}{t}(1-t)V$ by the assumption that V is an open neighborhood of 0. Since $b \in \overline{A}$, it follows that $b - \frac{1}{t}(1-t)V \cap A \neq \emptyset$ by the definition of closure, whence such a $d \in A$ exists. $\square$

Corollary: If $A \subseteq X$, then $\overline{\text{conv}}(A) = \overline{\text{conv}(A)}$.

Definition: A set $A \subseteq X$ is called balanced if for any $|\alpha| \leq 1$ and any $x \in A$, we have $\alpha x \in A$.

We say a set A is *absorbing* if for any $x \in X$, there is some $\varepsilon > 0$ such that $tx \in A$ for all $0 \leq t < \varepsilon$. We say A is absorbing at a if for any $x \in X$, there is $\varepsilon > 0$ such that $a + tx \in A$ for all $0 \leq t < \varepsilon$.

Given a seminorm p on V , the set $V = \{x : p(x) < 1\}$ is a convex balanced set that is absorbing at each of its points. We will show that the converse actually holds — that is, a vector space is a LCS if and only if the set of all open, convex, balanced subsets of X forms a neighborhood basis at 0.

Proposition: Suppose X is a vector space over F , and V is a nonempty convex balanced set that is absorbing at each of its points. Then, there is a unique seminorm p on X such that

$$V = \{x \in X : p(x) < 1\}.$$

Proof. Let

$$p(x) = \inf\{t \geq 0 : x \in tV\}.$$

Since V is absorbing, it follows that $X = \bigcup_{n=1}^{\infty} nV$, so this is a well-defined quantity. Furthermore, $p(0) = 0$.

To show that $p(\alpha x) = |\alpha|p(x)$, we suppose $\alpha \neq 0$, and since V is balanced,

$$\begin{aligned} p(\alpha x) &= \inf\{t \geq 0 : \alpha x \in tV\} \\ &= \inf\left\{t \geq 0 : x \in \frac{t}{\alpha}V\right\} \\ &= \inf\left\{t \geq 0 : x \in \frac{t}{|\alpha|}V\right\} \\ &= |\alpha| \inf\left\{\frac{t}{|\alpha|} : x \in \frac{t}{|\alpha|}V\right\} \\ &= |\alpha|p(x). \end{aligned}$$

We now observe that if $\alpha, \beta \geq 0$, then for any $a, b \in V$, we have

$$\begin{aligned} \alpha a + \beta b &= (\alpha + \beta) \left(\frac{\alpha}{\alpha + \beta} a + \frac{\beta}{\alpha + \beta} b \right) \\ &\in (\alpha + \beta)V, \end{aligned}$$

since V is convex. For any $x, y \in X$, take $p(x) = \alpha$ and $p(y) = \beta$, and let $\delta > 0$. Then, $x \in (\alpha + \delta)V$ and $y \in (\beta + \delta)V$ by the definition of the infimum, so that $x + y \in (\alpha + \beta + 2\delta)V$. Therefore, we have $p(x + y) \leq \alpha + \beta + 2\delta$, which holds for all $\delta > 0$, so that $p(x + y) \leq p(x) + p(y)$.

Finally, we will have to show that $V = \{x : p(x) < 1\}$. If $p(x) = \alpha < 1$, then for any $\alpha < \beta < 1$, we have $x \in \beta V \subseteq V$, which emerges from the fact that V is balanced. Thus, $V \supseteq \{x : p(x) < 1\}$. If $x \in V$, then $p(x) \leq 1$, and since V is absorbing at x , there is some $\varepsilon > 0$ such that for $0 < t < \varepsilon$, we have $y = x + tx \in V$. Yet, this means $x = \frac{1}{1+t}y$ with $y \in V$, so that $p(x) = \frac{1}{1+t}p(y) \leq \frac{1}{1+t} < 1$. $\square$

The seminorm p is called the Minkowski function for V , or the *gauge* of V .

Proposition: Let X be a TVS, and let $\mathcal{U}$ be the collection of open, convex, balanced subsets of X . Then, X is locally convex if and only if $\mathcal{U}$ is a basis for the neighborhood system at 0.

Proof. Suppose X is locally convex with topology defined by the family of seminorms $\mathcal{P}$. Then, we observe that each of the subbasic open neighborhoods of 0,

$$\{U(0, p, \varepsilon) : \varepsilon > 0, p \in \mathcal{P}\},$$

are open, convex, and balanced; since the set of all finite intersections of all these subbasic open neighbor-

hoods yields a neighborhood basis for 0, and finite intersections of open, balanced, convex subsets are open, balanced, and convex, so it follows that $\mathcal{U}$ contains the set of all finite intersections of all these subbasic open neighborhoods, whence $\mathcal{U}$ contains a basis for the neighborhood system at 0. In particular, this means $\mathcal{U}$ also must generate the neighborhood system at 0, so $\mathcal{U}$ forms a basis for the neighborhood system at 0.

Now, suppose $\mathcal{U}$ forms a basis for the neighborhood system at 0. Observe that an open, convex, balanced subset of V contains 0, and is itself absorbing. We claim that the family of Minkowski functionals

$$\mathcal{P} = \{p_V(\cdot - z) : z \in X, V \in \mathcal{U}\}$$

generates the topology for X . □

Thus, it follows that in a locally convex space, a net $(x_i)_i$ converges in the topology to x if and only if for all seminorms $p \in \mathcal{P}$, we have $p(x_i - x) \rightarrow 0$.

Example:

- (i) Suppose X is a locally compact Hausdorff space (or any Tychonoff space, but we will work with these for uniformity), and let $C(X)$ be the space of all continuous functions on X . Then, the family of seminorms $p_K(f) = \sup_{x \in K} |f(x)|$ yields a locally convex space where convergence is defined as uniform convergence on compact subsets.
- (ii) If X is a normed space, then the family of seminorms $p_\varphi(x) = |\varphi(x)|$ for each $\varphi \in X^*$ defines the weak topology on X .
- (iii) If X is a normed space, then the family of seminorms $p_x(\varphi) = |\varphi(x)|$ for each $x \in X$ defines the weak*, or w^* , topology on X^* .

The weak and weak* topologies are very important, and we will discuss them more in depth.

Metrizability and Normability

When we look at a locally convex space, we concern ourselves with whether the topology of the space is given by a metric or not. It turns out this is (effectively) equivalent to there being countably many seminorms defining the topology.

Proposition: Let $\{p_n : n \geq 1\}$ be a sequence of separating seminorms on X . For any $x, y \in X$, we may take

$$d(x, y) = \sum_{n=1}^{\infty} 2^{-n} \frac{p_n(x - y)}{1 + p_n(x - y)}.$$

Then, the topology on X is defined by this metric.

Definition: A Fréchet space is a locally convex space whose topology is defined by a metric that is complete.

Definition: If X is a topological vector space, and $B \subseteq X$, then we say B is bounded if, for every open set U containing 0, there is some $\varepsilon > 0$ such that $\varepsilon B \subseteq U$.

Proposition: If X is a locally convex space, then X is normable if and only if X has a nonempty bounded open set.

Proof. The open unit ball U_X is bounded. Therefore, we only need to show one direction. Suppose X is a locally convex space with a bounded open set U . Translating if necessary, we may assume $0 \in U$.

Since X is locally convex, there is an open, convex, balanced subset of U which we call V ; using the gauge of V , we may find a continuous seminorm p such that $\{x : p(x) < 1\} = V$. We will show that p is a norm and that p defines the topology on X .

First, let $0 \neq x \in X$. Let W_0 and W_x be disjoint open subsets with $0 \in W_0$ and $x \in W_x$. Since W_0 is absorbing, there is some $\varepsilon > 0$ such that $W_0 \supseteq \varepsilon U \supseteq \varepsilon V$. We note that $\varepsilon V = \{y : p(y) < \varepsilon\}$. Since $x \notin W_0$, we have $p(x) \geq \varepsilon$, so that p is a norm.

To show that the topology on X is generated by p , it suffices to show that if q is any continuous seminorm,

then there is some α such that $q \leq \alpha p$. Since q is continuous, it follows that there is some $\varepsilon > 0$ such that $\{x : q(x) < 1\} \supseteq \varepsilon U \supseteq \varepsilon V$. Yet, this means that if $p(x) < \varepsilon$, then $q(x) < 1$, whence $q \leq \varepsilon^{-1} p$. $\square$

Hyperplane Separation and Continuous Linear Functionals

We recall the classification of continuous linear functionals in a locally convex topological vector space.

Theorem: If X is a locally convex topological vector space with defining seminorms $\mathcal{P}$, and $f : X \rightarrow \mathbb{C}$ is a linear functional, then the following statements are equivalent:

- (i) f is continuous;
- (ii) f is continuous at 0;
- (iii) f is continuous at some point;
- (iv) $\ker(f)$ is closed;
- (v) there are $p_1, \dots, p_n \in \mathcal{P}$ and some constant C such that $|f(x)| \leq C \max\{p_1(x), \dots, p_n(x)\}$.

To understand hyperplane separation, we start with the most basic case — namely, that of separating a convex subset from the origin via a continuous linear functional.

Theorem: If X is a topological vector space, and G is an open convex nonempty subset of X that does not contain the origin, then there is a closed hyperplane M such that $M \cap G = \emptyset$.

Proof. We start by assuming that X is an $\mathbb{R}$ -linear space. Choose $x_0 \in G$, and let $H = x_0 - G$. Then, H is an open, convex subset of X that contains 0. Taking the gauge of H yields a continuous sublinear functional $q : X \rightarrow \mathbb{R}$ with $H = \{x : q(x) < 1\}$. Furthermore, since $x_0 \notin H$, we have that $q(x_0) \geq 1$.

Consider the subspace $Y = \{\alpha x_0 : \alpha \in \mathbb{R}\}$. Let $f_0 : Y \rightarrow \mathbb{R}$ be defined by taking $f_0(\alpha x_0) = \alpha q(x_0)$. Observe that if $\alpha \geq 0$, then $f_0(\alpha x_0) = q(\alpha x_0)$, and if $\alpha < 0$, then $f_0(\alpha x_0) = \alpha q(x_0) \leq \alpha < 0 \leq q(\alpha x_0)$, whence $f_0 \leq q$ on Y . Using the Hahn–Banach–Minkowski extension yields some $f : X \rightarrow \mathbb{R}$ a continuous linear functional such that $f|_Y = f_0$ and $f \leq q$ on X . Then, $M = \ker(f)$ is the desired hyperplane.

If $x \in G$, then $x_0 - x \in H$, so $f(x_0) - f(x) = f(x_0 - x) \leq q(x_0 - x) < 1$, while $f(x) > f(x_0) - 1 = q(x_0) - 1 \geq 0$ for all $x \in G$, whence $M \cap G = \emptyset$.

If X is a $\mathbb{C}$ -linear space, then since X is also an $\mathbb{R}$ -linear space, we obtain a continuous $\mathbb{R}$ -linear functional $f : X \rightarrow \mathbb{R}$ with $G \cap \ker(f) = \emptyset$. If $F(x) = f(x) - i f(ix)$, then F is a $\mathbb{C}$ -linear functional with $f = \operatorname{Re}(F)$ with $F(x) = 0$ if and only if $f(x) = f(ix) = 0$, so that $M = \ker(F)$ is the desired closed hyperplane. $\square$

Affine hyperplanes in X are given by $M' = M + x_0$ for some $x_0 \in X$ and hyperplane M . It turns out that we can extend any separating affine (closed) subspace of a topological vector space to an affine hyperplane.

Corollary: Let X be a topological vector space, and let G be an open, convex, nonempty subset of X . If Y is an affine subspace of X with $Y \cap G = \emptyset$, then there is a closed affine hyperplane M in X such that $Y \subseteq M$ and $M \cap G = \emptyset$.

Proof. Choosing some $x_0 \in Y$, we may consider $G - x_0$ and $Y - x_0$, so that Y may be assumed to be a subspace of X . Let $Q : X \rightarrow X/Y$ be the quotient map. Notice that $Q^{-1}(Q(G)) = \{y + G : y \in Y\}$, so that $Q(G)$ is open in X/Y . Furthermore, $Q(G)$ is also convex since Q is a linear map. Furthermore, since $Y \cap G = \emptyset$, $0 \notin Q(G)$. In particular, there is then a closed hyperplane N in X/Y such that $N \cap Q(G) = \emptyset$. Setting $M = Q^{-1}(N)$, we have that M is our desired hyperplane. $\square$

In the special case of a real topological vector space, it's useful to note that $X \setminus \ker(f)$ for a continuous linear functional $f : X \rightarrow \mathbb{R}$ has two connected components. This yields the following notion.

Definition: Let X be a real topological vector space. A subset $S \subseteq X$ is called an open half space if there is some continuous linear functional $f : X \rightarrow \mathbb{R}$ such that $S = f^{-1}((\alpha, \infty))$. We call S a closed half-space if there is a continuous linear functional $f : X \rightarrow \mathbb{R}$ such that $S = f^{-1}([\alpha, \infty))$.

We say A and B are strictly separated if they are contained in disjoint open half-spaces, while they are separated if they are contained in two closed half-spaces whose intersection is of the form $M = f^{-1}(\{\alpha\})$. That is, their intersection is a closed affine hyperplane. Notice also that continuous linear functionals are open maps, so that the interior of a closed half-space is an open half-space, and the closure of an open

half-space is a closed half-space.

Theorem (General Hyperplane Separation): Let X be a real topological vector space, and A and B disjoint convex subsets with A open. Then, there is a continuous linear functional $f: X \rightarrow \mathbb{R}$ and a real scalar α such that $A \subseteq f^{-1}((-\infty, \alpha))$ and $B \subseteq f^{-1}([\alpha, \infty))$.

Proof. Let $G = A - B$. Then, G is convex, and $G = \bigcup_{b \in B} (A - b)$, meaning G is open. Since $A \cap B = \emptyset$, it follows that $0 \notin G$. Thus, there is a closed hyperplane M in X such that $M \cap G = \emptyset$.

Let $f: X \rightarrow \mathbb{R}$ be a linear functional such that $M = \ker(f)$. Notice that $f(G)$ is a convex subset of $\mathbb{R}$ with $0 \notin f(G)$. Either, $f(x) > 0$ for all $x \in G$ or $f(x) < 0$ for all $x \in G$. Without loss of generality, we may assume that $f(x) > 0$ for all $x \in G$, so that $f(b) < f(a)$ for any $a \in A$ and $b \in B$, meaning there is some real number α such that $\sup_{b \in B} f(b) \leq \alpha \leq \inf_{a \in A} f(a)$. Since $f(A)$ and $f(B)$ are open intervals if B is open, it follows that $f < \alpha$ on B and $f > \alpha$ on A . $\square$

Lemma: If X is a topological vector space, K is a compact subset of X , and V is an open subset satisfying $K \subseteq V$, then there is an open neighborhood U of 0 such that $K + U \subseteq V$.

Proof. Let $\mathcal{O}_0$ denote the system of open neighborhoods of 0 . Suppose toward contradiction that for each $U \in \mathcal{O}_0$ that $K + U$ is not contained in V . We may choose vectors $x_U \in K$ and $y_U \in U$ such that $x_U + y_U \in V^c$. This yields a net $(y_U)_{U \in \mathcal{O}_0}$ and $(x_U)_{U \in \mathcal{O}_0}$ with $y_U \rightarrow 0$ and a subnet $x_{U'} \rightarrow x \in K$. Yet, this yields $(x_{U'} + y_{U'}) \rightarrow x$, implying that $x \in \overline{V^c} = V^c$, which yields a contradiction. $\square$

Theorem (Hyperplane Separation for Locally Convex Spaces): Let X be a real locally convex topological vector space, and let A and B be two disjoint closed convex subsets of X with B compact. Then, A and B are contained in disjoint open half-spaces.

Proof. Notice that B is a compact subset of A^c . Therefore, there is an open neighborhood U_1 of 0 such that $B + U_1 \subseteq A^c$. Since X is locally convex, we may find a continuous seminorm p on X such that $\{x : p(x) < 1\} \subseteq U_1$.

Define $U = \{x : p(x) < 1/2\}$. Then, $(B + U) \cap (A + U) = \emptyset$, and $A + U$ and $B + U$ are open convex subsets of X that contain A and B . Therefore, the result follows from the general hyperplane separation. $\square$

Rephrasing this theorem for locally convex spaces by using the fact that any linear functional φ over $\mathbb{C}$ is given by $\varphi(x) = \operatorname{Re}(\varphi(x)) - i \operatorname{Re}(\varphi(ix))$ yields the following.

Theorem (Hyperplane Separation for Locally Convex Spaces, Encore): Let X be a locally convex space, and let A and B be disjoint closed convex subsets of X with B compact. Then, there is a continuous linear functional $f: X \rightarrow \mathbb{C}$, some $\alpha \in \mathbb{R}$, and some $\varepsilon > 0$ such that for all $a \in A$ and $b \in B$, we have

$$\operatorname{Re}(f(a)) \leq \alpha < \alpha + \varepsilon \leq \operatorname{Re}(f(b)).$$

We will now discuss the dual space of one particular locally convex space.

Proposition: Let X be a locally compact Hausdorff space, and let $C(X)$ be topologized by uniform convergence on compact subsets. If $L: C(X) \rightarrow \mathbb{F}$ is a continuous linear functional, then there is a compact set K and a regular (complex) Borel measure μ on K such that $L(f) = \int_K f d\mu$ for any $f \in C(X)$. Conversely, all these measures define elements of $C(X)^*$.

Proof. Any compactly support measure defines an element of $C(X)^*$, and satisfies $|L(f)| \leq \|\mu\| \sup_{x \in K} |f(x)|$ for any such measure μ .

Conversely, let $L \in C(X)^*$. By the classification of continuous linear functionals, there are compact sets $K_1, \dots, K_n$ and a constant C such that $|L(f)| \leq C \max\{p_{K_j}(f) : j = 1, \dots, n\}$. Let $K = \bigcup_{j=1}^n K_j$. Then, $|L(f)| \leq \alpha p_K(f)$. In particular, for any $f \in C(X)$ with $f|_K = 0$, then $L(f) = 0$.

Define a linear functional $F: C(K) \rightarrow \mathbb{F}$ as follows. For any $g \in C(K)$, let $\tilde{g}$ be a continuous extension to X , and set $F(g) = L(\tilde{g})$. Suppose $\tilde{g}_1$ and $\tilde{g}_2$ are any such extensions; then, $\tilde{g}_1 - \tilde{g}_2 = 0$ on K , so that $L(\tilde{g}_1) = L(\tilde{g}_2)$, so that F is well-defined. Furthermore, F is linear. Since, for any $g \in C(K)$ and extension $\tilde{g} \in C(X)$, we have $|F(g)| \leq C p_K(g) = C \|g\|$. In particular, this means F is continuous, so there is some regular complex Borel measure μ in $M(K)$ such that $F(g) = \int g d\mu$. Notice that if $f \in C(X)$ is such that $f|_K = g \in C(K)$,

it follows that $L(f) = F(g) = \int f d\mu$. □

Weak Topologies and Extremal Structure

C^* -Algebras

Spectral Theory

Spectral theory is primarily concerned with representations of abelian C^* -algebras. Consider a normal operator N on H . Then, $C^*(N)$ is an abelian C^* -algebra, and the continuous functional calculus $f \mapsto f(N)$ yields a $*$ -isomorphism of $C(\sigma(N))$ onto $C^*(N)$. In particular, $f \mapsto f(N)$ is a representation of the abelian C^* -algebra $C^*(N)$.

We will find a way to extend this further, as well as find some useful applications to the structure of abelian von Neumann algebras and understanding unitary invariants for normal operators.

Definition: Let X be a set, Σ a σ -algebra on X , and H a Hilbert space. A *spectral measure* for (X, Σ, H) is a map $E: \Sigma \rightarrow B(H)$ satisfying

- (a) for each $\Delta \in \Sigma$, $E(\Delta)$ is a projection;
- (b) $E(\emptyset) = 0$ and $E(X) = 1$;
- (c) $E(\Delta_1 \cap \Delta_2) = E(\Delta_1)E(\Delta_2)$ for any $\Delta_1, \Delta_2 \in \Sigma$;
- (d) given pairwise disjoint $\{\Delta_n\}_{n=1}^\infty$, we have

$$E\left(\bigcup_{n=1}^\infty \Delta_n\right) = \sum_{n=1}^\infty E(\Delta_n),$$

where the latter is an SOT-convergent sum of pairwise orthogonal projections converging to the projection onto the supremum $\bigwedge_{n=1}^\infty E(\Delta_n)H$.

Recall that the SOT and WOT are defined by the seminorms

- $p_h(A) = \|Ah\|$ for the SOT, and
- $p_{h,k}(A) = |\langle Ah, k \rangle|$ for the WOT.

Example: If (X, Ω, μ) is a σ -finite measure space, and $\phi \in L^\infty(X, \mu)$, then $M_\phi \in B(L^2(X, \mu))$ can be defined by $M_\phi(f) = \phi f$.

A net $(\phi_i)_{i \in I}$ in $L^\infty(X, \mu)$ is w^* -convergent to ϕ if and only if $M_{\phi_i} \rightarrow M_\phi$ in WOT. This follows from the fact that if $f, g \in L^2(X, \mu)$, then $f\bar{g} \in L^1(X, \mu)$, where we have $L^1(X, \mu)^* = L^\infty(X, \mu)$, so that if $(\phi_i)_i \rightarrow \phi$ weak*, then

$$\begin{aligned} \langle M_{\phi_i} f, g \rangle &= \int \phi_i(f\bar{g}) d\mu \\ &\rightarrow \int \phi f\bar{g} d\mu \\ &= \langle M_\phi f, g \rangle, \end{aligned}$$

while the converse emerges from the fact that any element $f \in L^1(X, \mu)$ can be written as $f = g_1 \bar{g}_2$.

Example: If E is a spectral measure for (X, Ω, H) , then $E^{(n)}(\Delta) = E(\Delta)^{(n)}$ (under the diagonal embedding) is a spectral measure for $(X, \Omega, H^{(n)})$.

We will now use spectral measures to define a representation of a C^* -algebra.

Lemma: Let E be a spectral measure on (X, Ω, H) , and let $g, h \in H$. Then,

$$E_{g,h}(\Delta) = \langle E(\Delta)g, h \rangle$$

defines a countably additive measure on Ω with total variation bounded above by $\|g\|\|h\|$.

Proof. For countable additivity, we note the classic result that a sequence of projections $(P_n)_n \rightarrow P$ converges in WOT if and only if it converges in SOT. In particular, this means that if $\{\Delta_n\}_{n \in \mathbb{N}}$ is a sequence of disjoint subsets of X , then we have a sequence of projections

$$\left(\sum_{k=0}^n E(\Delta_k) \right)_n \rightarrow \sum_{k=0}^{\infty} E(\Delta_k)$$

converging in SOT, hence converging in WOT. Yet, this means that

$$\begin{aligned} \left\langle \left(\sum_{k=0}^n E(\Delta_k) \right) g, h \right\rangle &= \sum_{k=0}^n \langle E(\Delta_k)g, h \rangle \\ &\rightarrow \left\langle \left(\sum_{k=0}^{\infty} E(\Delta_k) \right) g, h \right\rangle, \end{aligned}$$

so that

$$\sum_{k=0}^{\infty} \langle E(\Delta_k)g, h \rangle = \left\langle E\left(\bigcup_{k=0}^{\infty} \Delta_k \right) g, h \right\rangle,$$

yielding our desired result.

If we have a finite set $\Delta_1, \dots, \Delta_n$ of pairwise disjoint sets, we choose $|\alpha_j| = 1$ such that $\left| \langle E(\Delta_j)g, h \rangle \right| = \alpha_j \langle E(\Delta_j)g, h \rangle$, whence

$$\begin{aligned} \sum_{j=1}^n |\mu(\Delta_j)| &= \left\langle \sum_{j=1}^n \alpha_j E(\Delta_j)g, h \right\rangle \\ &\leq \left\| \sum_{j=1}^n E(\Delta_j) \alpha_j g \right\| \|h\|. \end{aligned}$$

Notice that the family of $E(\Delta_j)$ are pairwise orthogonal projections, meaning that by Pythagoras's Theorem and the additivity of the spectral measure, we have

$$\begin{aligned} \left\| \sum_{j=1}^n E(\Delta_j) \alpha_j g \right\|^2 &= \sum_{j=1}^n \|E(\Delta_j)g\|^2 \\ &= \left\| E\left(\bigcup_{j=1}^n \Delta_j \right) g \right\|^2 \\ &\leq \|g\|^2, \end{aligned}$$

whence $\sum_{j=1}^n |\mu(\Delta_j)| \leq \|g\|\|h\|$. Since this holds for any finite collection of subsets, we have $\|\mu\|_{\text{TV}} \leq \|g\|\|h\|$. $\square$

We next concern ourselves with establishing “integration” with respect to the spectral measure.

Proposition: Let E be a spectral measure for (X, Ω, H) , and let $\phi: X \rightarrow \mathbb{C}$ be a bounded Ω -measurable function. Then, there is a unique operator $A \in B(H)$ such that for any $\varepsilon > 0$ and partition $\{\Delta_1, \dots, \Delta_n\}$ with $\sup_{x, x' \in \Delta_k} |\phi(x) - \phi(x')| < \varepsilon$ for each k , we have

$$\left\| A - \sum_{k=1}^n \phi(x_k) E(\Delta_k) \right\| < \varepsilon$$

for any $x_k \in \Delta_k$.

Proof. For each $g, h \in H$, define a (bounded) sesquilinear form

$$B(g, h) = \int \phi dE_{g, h},$$

with $|B(g, h)| \leq \|\phi\|_{L^\infty} \|g\| \|h\|$. Then, by the structure of bounded sesquilinear forms, there is a unique operator A such that $B(g, h) = \langle Ag, h \rangle$ for all $g, h \in H$.

Consider a partition $\{\Delta_1, \dots, \Delta_n\}$ satisfying the conditions in the proposition. For any arbitrary vectors $g, h \in H$ and $x_k \in \Delta_k$, we have

$$\begin{aligned} \left| \langle Ag, h \rangle - \sum_{k=1}^n \phi(x_k) \langle E(\Delta_k)g, h \rangle \right| &= \left| \sum_{k=1}^n \int_{\Delta_k} (\phi(x) - \phi(x_k)) dE_{g, h} \right| \\ &\leq \sum_{k=1}^n \int_{\Delta_k} |\phi(x) - \phi(x_k)| d|E_{g, h}| \\ &\leq \varepsilon \int d|E_{g, h}| \\ &\leq \varepsilon \|g\| \|h\|. \end{aligned}$$

This yields the desired result. □

For the operator A in the above statement, we write

$$A = \int \phi dE.$$

That is, for any $g, h \in H$ and bounded Ω -measurable function ϕ on X , we have

$$\left\langle \left(\int \phi dE \right) g, h \right\rangle = \int \phi dE_{g, h}.$$

If $B_\infty(X, \Omega)$ denotes the set of bounded Ω -measurable functions $\phi: X \rightarrow \mathbb{C}$ with the norm $\|\phi\| = \sup_{x \in X} |\phi(x)|$, then $B_\infty(X, \Omega)$ is a unital Banach algebra — in fact, it is an abelian C^* -algebra.

Proposition: Suppose E is a spectral measure for (X, Ω, H) , and $\rho: B_\infty(X, \Omega) \rightarrow B(H)$ is defined by taking $\rho(\phi) = \int \phi dE$.

Then, ρ is a representation of $B_\infty(X, \Omega)$ and $\rho(\phi)$ is a normal operator for every $\phi \in B_\infty(X, \Omega)$.

Proof. We will only focus on showing that ρ is multiplicative. Suppose $\phi, \psi \in B_\infty(X, \Omega)$. Let $\varepsilon > 0$, and let $\{\Delta_1, \dots, \Delta_n\}$ be a Borel partition of X satisfying

$$\sup \{ |\omega(x) - \omega(x')| : x, x' \in \Delta_k, \omega = \phi, \psi, \phi\psi \} < \varepsilon.$$

In particular, we have that for any $x_k \in \Delta_k$ and $\omega = \phi, \psi, \phi\psi$, we have

$$\left\| \int \omega dE - \sum_{k=1}^n \omega(x_k) E(\Delta_k) \right\| < \varepsilon.$$

The triangle inequality then gives

$$\left\| \int \phi \psi dE - \left(\int \phi dE \right) \left(\int \psi dE \right) \right\| \leq \varepsilon + \left\| \sum_{k=1}^n \phi(x_k) \psi(x_k) E(\Delta_k) - \left(\sum_{i=1}^n \phi(x_i) E(\Delta_i) \right) \left(\sum_{j=1}^n \psi(x_j) E(\Delta_{x_j}) \right) \right\|$$

$$+ \left\| \left(\sum_{i=1}^n \phi(x_i) E(\Delta_i) \right) \left(\sum_{j=1}^n \psi(x_j) E(\Delta_j) \right) - \left(\int \phi dE \right) \left(\int \psi dE \right) \right\|.$$

Since $E(\Delta_i)E(\Delta_j) = E(\Delta_i \cap \Delta_j)$, and $\{\Delta_1, \dots, \Delta_n\}$ is a partition, we have that the second term on the right-hand side is zero. Thus, we have

$$\left\| \int \phi \psi dE - \left(\int \phi dE \right) \left(\int \psi dE \right) \right\| \leq \varepsilon + \left\| \left(\sum_{i=1}^n \phi(x_i) E(\Delta_i) \right) \left(\sum_{j=1}^n \psi(x_j) E(\Delta_j) - \int \psi dE \right) \right\|$$

$$+ \left\| \left(\sum_{i=1}^n \phi(x_i) E(\Delta_i) - \int \phi dE \right) \left(\int \psi dE \right) \right\|$$

$$\leq \varepsilon (1 + \|\phi\| + \|\psi\|).$$

Thus, we get multiplicativity. $\square$

Corollary: If X is a compact Hausdorff space, and E is a spectral measure on H defined on the Borel sets of X , then $\rho: C(X) \rightarrow B(H)$ given by $\rho(u) = \int u dE$ is a representation of $C(X)$.

Thus far, all we have obtained is a variety of the properties of a spectral measure. However, we do not know if such a measure exists. The following is the proof that it does.

Theorem: Suppose $\rho: C(X) \rightarrow B(H)$ is a representation. Then, there is a unique spectral measure E defined on the Borel subsets of X such that for all $g, h \in H$, $E_{g,h}$ is a regular measure, and we have

$$\rho(u) = \int u dE$$

for all $u \in C(X)$.

Theorem: Let $g, h \in H$. The map $u \mapsto \langle \rho(u)g, h \rangle$ is a linear functional on $C(X)$ with norm at most $\|g\| \|h\|$. So, there is a unique measure $\mu_{g,h}$ in $M(X)$ such that

$$\langle \rho(u)g, h \rangle = \int u d\mu_{g,h}.$$

The pairing $(g, h) \mapsto \mu_{g,h}$ is sesquilinear, and for fixed $\phi \in B_\infty(X)$, the form

$$[g, h] = \int \phi d\mu_{g,h}$$

is a sesquilinear form with $\|[g, h]\| \leq \|\phi\| \|g\| \|h\|$. Therefore, there is a unique bounded operator A such that $[g, h] = \langle Ag, h \rangle$ with $\|A\| \leq \|\phi\|$. We will denote this operator A by $\tilde{\rho}(\phi)$. We obtain a well-defined map $\tilde{\rho}: B_\infty(X) \rightarrow B(H)$ satisfying $\|\tilde{\rho}(\phi)\| \leq \|\phi\|$ such that

$$\langle \tilde{\rho}(\phi)g, h \rangle = \int \phi d\mu_{g,h}.$$

We start by claiming that $\tilde{\rho}: B_\infty(X) \rightarrow B(H)$ is a representation with $\tilde{\rho}|_{C(X)} = \rho$. That $\tilde{\rho}$ restricted to $C(X)$ is equal to ρ follows directly from the definition of $\tilde{\rho}$.

Now, let $\phi \in B_\infty(X)$. Consider ϕ as an element of $M(X)^* = C(X)^{**}$ by viewing it as corresponding to the linear functional

$$\mu \mapsto \int \phi d\mu.$$

By the Banach–Alaoglu theorem, we have that $\{u \in C(X) : \|u\| \leq \|\phi\|\}$ is weak*-dense in $\{L \in M(X)^* : \|L\| \leq \|\phi\|\}$. Then, there is a net $(u_i)_i \subseteq C(X)$ with $\|u_i\| \leq \|\phi\|$ for all i and

$$\int \phi d\mu = \lim_i \int u_i d\mu$$

for all $\mu \in M(X)$. For any $\psi \in B_\infty(X)$, we have $\psi\mu \in M(X)$ for any $\mu \in M(X)$. In particular, we have

$$\int \phi \psi d\mu = \lim_i \int u_i \psi d\mu$$

for all $\psi \in B_\infty(X)$ and $\mu \in M(X)$, whence $\widetilde{\rho}(u_i \psi) \rightarrow \widetilde{\rho}(\phi \psi)$ in WOT. For any $\psi \in C(X)$, we have that

$$\begin{aligned} \widetilde{\rho}(\phi \psi) &= \lim_i \widetilde{\rho}(u_i \psi) \\ &= \lim_i \rho(u_i) \rho(\psi) \\ &= \widetilde{\rho}(\phi) \rho(\psi). \end{aligned}$$

In particular, we have $\widetilde{\rho}(u_i \psi) = \rho(u_i) \widetilde{\rho}(\psi)$ for any $\psi \in B_\infty(X)$ for all $u_i \in C(X)$. Therefore, we have $\widetilde{\rho}(u_i \psi) \rightarrow \widetilde{\rho}(\phi \psi)$ in WOT, whence

$$\widetilde{\rho}(\phi \psi) = \widetilde{\rho}(\phi) \widetilde{\rho}(\psi)$$

for any $\phi, \psi \in B_\infty(X)$.

The linearity of measures gives that $\widetilde{\rho}$ itself is linear. A similar bootstrapping argument gives $\widetilde{\rho}(\phi)^* = \widetilde{\rho}(\overline{\phi})$.

Now, let $\Delta \subseteq X$ be Borel. Let $E(\Delta) = \widetilde{\rho}(\chi_\Delta)$. We will show that E is a spectral measure.

First, since χ_Δ is a hermitian idempotent in $B_\infty(X)$, it follows that $E(\Delta)$ is a projection by the classification of projections on a Hilbert space. Furthermore, $\chi_\emptyset = 0$ and $\chi_X = 1$, so that $E(\emptyset) = 0$ and $E(X) = 1$. Since

$$\begin{aligned} E(\Delta_1 \cap \Delta_2) &= \widetilde{\rho}(\chi_{\Delta_1 \cap \Delta_2}) \\ &= \widetilde{\rho}(\chi_{\Delta_1} \chi_{\Delta_2}) \\ &= E(\Delta_1) E(\Delta_2). \end{aligned}$$

Finally, let $\{\Delta_n : n \geq 0\}$ be a pairwise disjoint sequence of Borel sets. Then, for fixed $h \in H$ and $\Lambda_n = \bigcup_{k=n+1}^\infty \Delta_k$, we have

$$\begin{aligned} \left\| E\left(\bigcup_{k=1}^\infty \Delta_k\right) - \sum_{k=1}^n E(\Delta_k) \right\|^2 &= \langle E(\Lambda_n) h, E(\Lambda_n) h \rangle \\ &= \langle E(\Lambda_n) h, h \rangle \\ &= \int \chi_{\Lambda_n} d\mu_{h,h} \\ &= \sum_{k=n+1}^\infty \mu_{h,h}(\Delta_k) \\ &\xrightarrow{n \rightarrow \infty} 0. \end{aligned}$$

Thus, E is a spectral measure.

Finally, we must show that $\rho(u) = \int u \, dE$ for all $u \in C(X)$. In fact, we will show that $\tilde{\rho}(\phi) = \int \phi \, dE$ for all $\phi \in B_\infty(X)$. For a fixed $\phi \in B_\infty(X)$, we choose a Borel partition $\{\Delta_1, \dots, \Delta_n\}$ such that $\sup\{|\phi(x) - \phi(x')| : x, x' \in \Delta_k\} < \varepsilon$ for each k . Then,

$$\left\| \phi - \sum_{k=1}^n \phi(x_k) \chi_{\Delta_k} \right\| < \varepsilon$$

for any choices of $x_k \in \Delta_k$. Since $\|\tilde{\rho}\| = 1$, it follows that

$$\left\| \tilde{\rho}(\phi) - \sum_{k=1}^n \phi(x_k) E(\Delta_k) \right\| < \varepsilon,$$

so that $\tilde{\rho}(\phi) = \int \phi \, dE$ for any $\phi \in B_\infty(X)$.

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